COMPUTER ORGANIZATION AND ARCHITECTURE

Assignment - 1

Submitted By -

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Question 01. What is RASPBERRY Pi (5).?

Raspberry pi 5 is like a small computer and it is made on a single board. Single board computer means in normal computer motherboard, processor, RAM are separated. And in Raspberry Pi present in a single board. Raspberry Pi foundation developed Raspberry pi 5.
Size of raspberry pi is very small we can hold in hand also but work like a normal computer. In normal computer we are using different monitor, keyboard, mouse, internet similarly we can connect to all these in raspberry pi 5.
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Main Features

- To learn Programming
- Control electronic devices
- Built in Wi-fi and Bluetooth

Question 02. Explain the 10 Key Takeaways from an AI Summit.?

- AI should be secure and reliable.
- Better technology is needed for the increased use of AI.
- Every Country has different AI Rules
- More AI technology and computer need for growth.
- Students should learn AI skills for jobs.
- AI should be used in a safe way.
- Researchers help create new AI ideas.
- AI should solve real-life problems and improve people lives.
- AI is now being used in real life very much.
- AI also helped in increasing the economy

Question 03. What is NPTEL.?

The Full form of NPTEL is National Programme on Technology Enhanced Learning. It is an online learning platform started by IITs and IISc support of Government of India.

It provides free online courses on subjects like Computer Science, Engineering, Science, and many topics. Students also learn new skills.

Question 04. How do we decide which programming language to use for designing an application? What are the important parameters.?

Designing a programming language for an application depends on the type of application we want to make. Different languages are used for different purposes.
For example, Python is used for AI and data science, Java is used for Android apps, and JavaScript is used for websites.

Before choosing language we check important points:

1. Type of Application
2. Easy to Learn and Use
3. Speed and Performance
4. Security

Question 05. Explain RAM Size, Word Size, and the meaning of a 64-bit System.?

1. RAM Size

The full form of RAM is Random Access Memory. RAM is the temporary memory of a computer. It helps the computer run programs and applications.
RAM Size tells us how much data a computer can keep ready for use at one time.

Example: A computer with 8 GB RAM can handle more programs compared to a computer with 4 GB RAM.

2. Word Size

Word size means the amount of data a computer can process at one time.
Example: A 32-bit computer can process 32 bits of data at one time, and a 64-bit computer can process 64 bits of data at one time.

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3. Meaning of a 64-bit System

A 64-bit system is a type of computer system that can process 64 bits of data at one time.
It can handle more memory and bigger programs compared to a 32-bit system.

Question 06. Why is GPU faster than CPU.?

GPU (Graphics Processing Unit) is faster than CPU (Central Processing Unit) for some tasks because it can do many calculations at the same time.

A CPU has a few powerful cores so it does tasks one by one or a few at a time.
A GPU has many small cores so it can do thousands of small tasks together at the same time.

Question 07. Explain CPU, GPU, and TPU.?

Feature	CPU	GPU	TPU
Full Form	Central Processing Unit	Graphics Processing Unit	Tensor Processing Unit
Main Work	Runs normal computer tasks	Handles graphics and many calculations	Used for AI and Machine Learning tasks
Used In	Computers, laptops, mobiles	Games, videos, graphics	Google AI systems and deep learning

Question 08. Write about the Last Year's Turing Award.?

The Turing Award is one of the biggest awards in field of Computer Science.
It gives people who make important contribution to computer and technology.

The last year Turing Award was given to Andrew Barto and Richard Sutton for their important work in Reinforcement Learning.

Question 09. What is the Stampeding Herd Problem?

The Stampeding Herd Problem is a problem in computer science where many users or programs try to access the same resource at the same time.

Because of this, the system can become slow or overloaded.

Question 10. Who are the founders of Intel and AMD.?

Intel was founded by Robert Noyce and Gordon Moore in 1968.

AMD (Advanced Micro Devices) was founded by Jerry Sanders and his team in 1969.

Question 11. Which CPU organization is your laptop machine.?

My laptop has a 13th Generation Intel Core i5-1334U processor.

It uses x86-64 CPU architecture.

It can process 64 bits of data at one time.